

Lesson 1: Your Digital Footprint

1. What is a digital footprint?

- a) Online record of activities
- b) List of followers and likes
- c) Temporary browser cache
- d) Virus that tracks location

2. Which statement is true about online interactions?

- a) You can always trust people online
- b) Not everyone online is honest
- c) Online friends = real friends
- d) Never talk to strangers

3. Which two types of digital footprints exist?

- a) Good and Bad
- b) Affirmative and Dissenting
- c) Active and Passive
- d) Positive and Negative

4. Why be careful with strangers online?

- a) They're always trustworthy
- b) They may steal data
- c) They're usually real friends
- d) They give good advice

5. How can you protect your privacy online?

- a) Share personal info freely
- b) Keep passwords public
- c) Decide what to keep private
- d) Trust everyone online

6. What to do before posting about others?

- a) Share everything you think
- b) Avoid posting at all
- c) Only post positive things
- d) Post only with permission

7. Why be careful about online actions?

- a) It's fun to post anything
- b) Digital footprints are invisible
- c) Data can be misused
- d) Nobody sees online actions

8. What's a risk of posting travel plans?

- a) Thieves know your home is empty
- b) Make new friends
- c) Receive postcards
- d) Friends join the trip

9. Digital footprints are used by:

- a) Customers
- b) Clients
- c) Advertisers
- d) Shopkeepers

10. Why think before posting?

- a) It's fun to share everything
- b) Everyone else is doing it
- c) There are no consequences
- d) It affects your digital footprint and reputation

Answer:

1a	2b	3d	4b	5c	6d	7c	8a	9c	10d
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Lesson 2: Safety and Security: Online Profiles

1. What is an online profile?

- a) Digital picture of yourself
- b) A photo album
- c) A type of game
- d) Printed papers

2. Why keep your profile private?

- a) To stop strangers from seeing details
- b) To avoid random friend requests
- c) To make it professional
- d) All of the above

3. What do the 'lock' and 'world' icons mean?

- a) Lock = public, World = private
- b) Lock = private, World = public
- c) They show hobbies
- d) They mean nothing

4. What should you avoid posting?

- a) Passwords
- b) Holiday dates
- c) Photos of friends
- d) Hobbies

5. If someone makes you uncomfortable online, what first?

- a) Ignore them
- b) Block and report
- c) Share more info
- d) Meet in person

6. Why is sharing school name/phone risky?

- a) Strangers can misuse it
- b) It makes your profile boring
- c) It reduces followers
- d) It wastes space

7. Which shows strong online security?

- a) Using the same password everywhere
- b) Accepting strangers online
- c) Sharing personal details publicly
- d) Checking and managing privacy settings

8. Should you always use your real photo?

- a) Depends on choice & safety
- b) Yes, always
- c) No, never
- d) Only if famous

9. Why limit public sharing of posts?

- a) To look interesting
- b) To get more followers
- c) To protect personal info
- d) To be an internet celebrity

10. A post online means:

- a) A printed letter
- b) Info shared on websites/apps
- c) Your online identity
- d) A photo album

Answer:

1a	2d	3b	4a	5b	6a	7d	8a	9c	10b
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Lesson 3: Safety and Security Online Information

1. What is online shopping?

- a) Buying items on the internet
- b) Buying things with cash only
- c) Buying through a phone call
- d) Buying using a physical cart

2. Why check your bank statement after online shopping?

- a) To see discounts and offers
- b) To confirm your card works
- c) To spot wrong or fake charges
- d) To check if you're popular

3. What should you keep safe in online banking?

- a) Movie list
- b) Login and password
- c) Recipes
- d) Contacts

4. Which of the following is never used as an online payment method?

- a) Credit card
- b) Cash on delivery
- c) Bitcoin
- d) Barter system

5. Why avoid sharing your banking login?

- a) To make access easier for others
- b) To receive discounts
- c) To gain followers
- d) To keep your account safe

6. What to do if someone asks for your login?

- a) Share it with them
- b) Refuse or ignore
- c) Ask them for theirs
- d) Post it online

7. What does online banking let you do?

- a) Manage money online
- b) Order pizza
- c) Watch movies
- d) Play games

8. Online auctions are like:

- a) Cooking classes
- b) Online shopping
- c) Exams
- d) Movie theaters

9. Main caution in online auctions?

- e) Verify the seller's credibility
- f) Use a strong internet connection
- g) Bid at any time
- h) Focus only on low prices

10. Best safety practice for online shopping?

- a) Buy only from trusted sites
- b) Pay only with cash
- c) Share details with strangers
- d) Use weak passwords

Answer:

1a	2c	3b	4d	5d	6b	7a	8b	9d	10a
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Lesson 4: Safety and Security – Cybercrime

1. What is identity theft?

- a) Using strong passwords
- b) Stealing someone's personal data
- c) Sharing info safely
- d) None of the above

2. Which personal detail is least valuable to identity thieves?

- a) Bank account details
- b) Social Security number
- c) Name and address
- d) Favorite color

3. Why keep personal info private online?

- a) To avoid theft and fraud
- b) To avoid making friends
- c) To get more followers
- d) To win prizes

4. What to do with a suspicious email asking for info?

- a) Reply with details
- b) Click its links
- c) Delete it
- d) Forward it

5. What trick do thieves often use?

- a) Strong passwords
- b) Phishing emails
- c) Secure websites
- d) None of the above

6. Which is good password practice?

- a) Use “password123”
- b) Share with friends
- c) Mix letters, numbers, and symbols
- d) All of the above

7. Which UK agency fights cybercrime?

- a) Cybersecurity Authority
- b) Identity Protection Agency
- c) Action Fraud
- d) None of the above

8. What can identity theft cause?

- a) Reputation damage
- b) Financial loss
- c) Credit score problems
- d) All of the above

9. How to reduce risk of identity theft?

- a) Use weak passwords
- b) Check your accounts often
- c) Share data openly
- d) None of the above

10. How can identity theft happen online?

- a) Stolen wallets
- b) Phishing and hacking
- c) Credit card use
- d) None of the above

Answer:

1b	2d	3d	4c	5b	6c	7c	8d	9b	10b
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Lesson 5: Online Content Management

1. Which is NOT a sign of bad online content?

- a) Hate speech
- b) Violence
- c) Explicit language
- d) Trusted study materials

2. What should you do if you see bad content?

- a) Share with friends
- b) Report to a trusted adult
- c) Ignore and keep browsing
- d) Delete history

3. Who works to remove child abuse material online?

- a) CEOP (UK)
- b) Internet Watch Foundation (IWF)
- c) NCMEC (USA)
- d) Common Sense Media

4. How can you protect privacy online?

- a) Share info openly
- b) Use strong passwords
- c) Click unknown links
- d) None of the above

5. Why do some sites have age limits?

- a) To ensure content is suitable for the targeted audience
- b) To make sites popular
- c) To promote advertisements
- d) To increase website traffic

6. Exposure to bad content can make kids feel:

- a) Aggressive
- b) Joyful
- c) Fearful
- d) None of the above

7. Minimum age to open a YouTube account?

- a) 13 years
- b) 18 years
- c) 20 years
- d) No limit

8. What is a reliable source online?

- a) Hate speech
- b) Trusted websites and articles
- c) Explicit sites
- d) None of the above

9. What should you never share online?

- a) Educational sites
- b) Favorite color
- c) Interesting articles
- d) Personal information

10. If you see adult or explicit content, what should you do?

- a) Discuss online
- b) Ignore it
- c) Share with friends
- d) Report to a trusted adult

Answer:

1d	2b	3b	4b	5a	6c	7a	8b	9d	10d
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Lesson 6: Improving Your Digital Footprint

1. How can you improve your digital footprint?

- a) Post without thinking
- b) Share passwords with friends
- c) Be mindful of what you post
- d) Post negative comments

2. How should you treat others online?

- a) Be rude
- b) Ignore everyone
- c) Share others' info
- d) Be respectful and kind

3. Why be careful sharing personal info online?

- a) To protect your online identity
- b) To gain followers
- c) To make new friends
- d) To show off

4. Which is NOT a tip for a good profile?

- a) Avoid sharing personal info
- b) Be respectful
- c) Share offensive content
- d) Highlight achievements

5. What are cookies?

- a) Small text files saved by websites
- b) Sweet baked treats
- c) Large code files
- d) None of the above

6. Why care about your digital footprint?

- a) Helps your online reputation
- b) Helps make friends
- c) Only matters to employers
- d) No impact

7. How to make your social media more positive?

- a) Share personal info freely
- b) Post fake news
- c) Be respectful and positive
- d) Post negative comments

8. In a job context, what does ‘YOU ARE HIRED!’ mean?

- a) Don’t apply for jobs
- b) You got the job
- c) Not suitable for work
- d) Update profile

9. What do employers like to see online?

- a) Negative job comments
- b) Teamwork and collaboration
- c) Offensive jokes
- d) Frequent arguments

10. How to keep a positive footprint when posting?

- a) Post without thinking
- b) Join online arguments
- c) Think before posting
- d) Share info freely

Answer:

1c	2d	3a	4c	5a	6a	7c	8b	9b	10c
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Lesson 7: Keeping Yourself and Friends Safe Online

1. What does phishing mean in cybersecurity?

- a) Fake scam pretending to be real to steal info
- b) A hobby
- c) A game
- d) A virus

2. Main goal of malware?

- a) Protect your computer
- b) Help browse safely
- c) Damage computer or steal info
- d) Give free software

3. What to do if your computer has malware?

- a) Ignore it
- b) Share your computer
- c) Delete all files
- d) Run antivirus

4. How to protect from phishing?

- a) Click all links
- b) Open emails from strangers
- c) Avoid suspicious emails/links
- d) Share info by email

5. What is pharming?

- a) Fake sites stealing info
- b) Online plant growing
- c) Farming game
- d) Online shopping

6. Which malware encrypts files and asks ransom?

- a) Virus
- b) Worm
- c) Ransomware
- d) d) Spyware

7. What is a Trojan?

- a) A virus that spreads through email attachments
- b) Malware disguised as legitimate software
- c) A type of input device like mouse
- d) A physical component inside a computer

8. What is malware?

- a) Hardware
- b) Software that protects
- c) Harmful software that steals/damages
- d) A network

9. What is a worm?

- a) Improves performance
- b) Deletes all files
- c) Encrypts files
- d) Malware that spreads through networks

10. What does spyware do?

- a) Helps browse faster
- b) Collects info secretly
- c) Protects from viruses
- d) Organizes files

Answer:

1a	2c	3d	4c	5a	6c	7b	8c	9d	10b
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Lesson 8: Keeping Yourself and Your Friends Safe Online

- 1. What is the risk of downloading files from unknown sites?**
 - a) Malware/viruses
 - b) Faster speed
 - c) Extra storage
 - d) Better performance

- 2. What is it called when copyrighted content is used without permission?**
 - a) Copyright sharing
 - b) Copyright rule
 - c) Copyright legality
 - d) Copyright infringement

- 3. Which files are usually shared for fun?**
 - a) Movies, songs, clips
 - b) Safety videos
 - c) Code files
 - d) Spreadsheets

- 4. How does security software stop unsafe files?**
 - a) Detect & delete/quarantine
 - b) Block internet fully
 - c) Remove all downloads
 - d) Slow your PC

- 5. What is the main issue with sharing copyrighted music?**
 - a) Legal & ethical
 - b) Supports artists
 - c) Copyright breach
 - d) Helps the industry

6. Which of the following is an inappropriate reason for file sharing?

- a) Work purpose
- b) Hard-to-find content
- c) Illegal downloads
- d) Legal sharing

7. Which platform is legal for music?

- a) Spotify
- b) Pirate Bay
- c) Hotfile
- d) Rapid Share

8. What is the purpose of copyright?

- a) Block all access
- b) Support illegal sharing
- c) Stop digital use
- d) Protect creators' rights

9. What happens if you share copyrighted files illegally?

- a) Free downloads
- b) Fines/penalty
- c) More fame
- d) Special access

10. Which software is widely used in business for secure file sharing?

- a) Google Drive
- b) Rapid Share
- c) Spotify
- d) Pirate Bay

Answer:

1a	2d	3a	4a	5c	6c	7a	8d	9b	10a
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Lesson 9: Protecting Your Computer Against Viruses

1. How can viruses enter a computer?

- a) Just turning on
- b) Strong password
- c) Candy
- d) Infected USB

2. Why keep your PC updated?

- a) Fix security holes
- b) Add new games
- c) Organize files
- d) Look nicer

3. What kind of passwords are best?

- a) Weak/guessable
- b) Strong & unique
- c) Shareable
- d) None

4. Which email attachment is risky?

- a) From unknown sender
- b) From boss
- c) From antivirus
- d) From friend

5. What is the main task of antivirus software?

- a) Erase all files
- b) Make PC faster
- c) Scan & remove malware
- d) Build passwords

6. Why back up files often?

- a) Save storage
- b) Faster system
- c) Avoid loss if infected
- d) Delete unwanted data.

7. What is often missing in free antivirus?

- a) Password tool
- b) Quick scan
- c) Updates
- d) Firewall

8. How often should you scan your PC?

- a) Once a year
- b) Never
- c) At least monthly
- d) Only when emailed.

9. What is a "data safe" in antivirus?

- a) Vault for sensitive files
- b) Money locker
- c) Free storage
- d) Software downloader

10. What must you avoid when browsing?

- a) Clicking shady links
- b) Changing wallpaper
- c) Using antivirus
- d) Short passwords

Answer:

1d	2a	3b	4a	5c	6c	7d	8c	9a	10a
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Lesson 10: Staying Secure on Social Media

1. What improves social media security?

- a) Keep defaults
- b) Share password
- c) Post personal info
- d) Enable 2FA

2. What to do if someone misbehaves online?

- a) Ignore
- b) Copy chat
- c) Report/block
- d) Reply back

3. Why review privacy settings?

- a) Improve profile look
- b) Add more friends
- c) Control who sees info
- d) Share with strangers

4. Which password is weak?

- a) abcd1234
- b) Totoro@829
- c) He!!0@341
- d) \$mi!#9090

5. What is the purpose of two-factor authentication?

- a) Only you can log in
- b) Easy to hack
- c) Share with others
- d) Open to all

6. What does VPN stand for?

- a) Virtual Present Network
- b) Virtual Private Network
- c) Vertical Private Net
- d) Virtual Net Program

7. What is the main use of VPN?

- a) Boost storage
- b) Get more likes
- c) Hide online activity
- d) Speed up device

8. Which is not advised for social media security?

- a) Avoid using public Wi-Fi
- b) Use strong, unique passwords
- c) Enable two-factor authentication (2FA)
- d) Share your passwords over public Wi-Fi

9. Why create strong passwords?

- a) Easy guessing
- b) Extra security
- c) Same for all accounts
- d) Never forget

10. Which is good practice for social media passwords?

- a) Common phrases
- b) Same everywhere
- c) Unique & strong
- d) Share with family

Answer:

1d	2c	3c	4a	5a	6b	7c	8d	9b	10c
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Lesson 11: Staying Safe Online

1. What should you avoid sharing online?

- a) School name
- b) Hobbies
- c) Address/phone
- d) Positive notes

2. What are internet trolls?

- a) People posting hate for attention
- b) People giving advice
- c) People defending others
- d) People posting positive notes.

3. What is cyberbullying?

- a) Encouraging others
- b) Posting mean comments, threats, or abusive messages
- c) Sharing fun clips
- d) Posting positive content

4. What should you do if bullied online?

- a) Share it around
- b) Argue back
- c) Ignore always
- d) Report & block

5. Which is NOT cyberbullying?

- a) Motivation
- b) Trolling
- c) Outing
- d) Exclusion

6. What is the focus of online safety?

- a) Social media use
- b) Reporting harmful or suspicious activity
- c) Protecting personal information
- d) All of the above

7. How can you protect from trolls?

- a) Share details
- b) Argue online
- c) Think before posting, report
- d) Reply to all

8. How do internet trolls typically behave or feel?

- a) Think they're safe, no concern
- b) Proud
- c) Annoyed
- d) Sorry

9. How can you be cautious of strangers online?

- a) Accept all
- b) Avoid requests/chats with strangers
- c) Share personal info
- d) Chat in private

10. How can you stay safe online?

- a) Think before posting, keep privacy
- b) Share info with friends
- c) Argue online
- d) Reply to all

Answer:

1c	2a	3b	4d	5a	6d	7c	8a	9b	10a
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Lesson 12: Stay Secure Online

1. What is clickjacking?

- a) Track website clicks
- b) Trick to click unintended item
- c) Boost site traffic
- d) Faster clicking

2. What is the risk of location leakage?

- a) Fewer views
- b) Slower device
- c) Attackers get your data
- d) Battery drain

3. Why should you avoid using easy passwords?

- a) Extra security
- b) Fun to use
- c) Impress friends
- d) Easy recall.

4. What is the purpose of a secure network?

- a) Make it complex
- b) Slow payments
- c) Allow public Wi-Fi
- d) Protect sensitive info in transactions

5. Why might someone create a fake online profile?

- a) Pretend & trick friends
- b) Look professional
- c) Gain followers
- d) Play pranks.

6. What to do with suspicious links/attachments?

- a) Click to check
- b) Delete without opening
- c) Share with friends
- d) Open carefully

7. Full form of OSN?

- a) Online Social Network
- b) Optimal Social Nation
- c) Offline Social Net
- d) Optimal Service Net

8. What is the purpose of biometric measures?

- a) Share info
- b) Create strong passwords
- c) Identify by physical traits
- d) Argue online

9. Risk of sharing health info?

- a) Insurance impact via data leakage
- b) Connect with people
- c) Better ads
- d) Personal reminders.

10. How often should passwords be changed?

- a) After every login
- b) Only when you forget them
- c) Never, under any circumstances
- d) Periodically or when there is a security risk

Answer:

1b	2c	3a	4d	5a	6b	7a	8c	9a	10d
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Lesson 13: Understanding and Designing Algorithms

1. What is an algorithm?

- a) Virus
- b) Hardware
- c) Language
- d) Rules/instructions to solve a problem

2. Which algorithm finds a specific item?

- a) Searching
- b) Sorting
- c) Pathfinding
- d) Math

3. Why does order of steps matter?

- a) Doesn't matter
- b) Makes longer
- c) Affects result/efficiency
- d) Confuses

4. Which of the following is not a valid way to represent an algorithm?

- a) Music notes
- b) Flowchart
- c) Pseudocode
- d) Programming

5. What is pseudocode used for?

- a) Drawing charts
- b) Step-by-step in simple form
- c) Writing programs
- d) Math solving

6. What is the purpose of pseudocode?

- a) Clear description of algorithm
- b) Artistic design
- c) Code writing
- d) Visual display

7. Which command calculates in an algorithm?

- a) READ
- b) PRINT
- c) COMPUTE
- d) GET

8. What is the primary purpose of an algorithm?

- a) Solve problems by steps
- b) Perform math only
- c) Confuse users
- d) Create virus

9. What does path-finding algorithm find?

- a) Longest path
- b) Most complex path
- c) Path with many stops
- d) Shortest or best path

10. Which of the following describes a good input in an algorithm?

- a) Ambiguous
- b) Well-defined
- c) Not specified
- d) Optional

Answer:

1d	2a	3c	4a	5b	6a	7c	8a	9d	10b
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Lesson 14: Decision Making in Algorithms

- 1. What is the result of a decision in an algorithm?**
 - a) Numbers
 - b) True/False
 - c) Colors
 - d) Symbols

- 2. Which shape shows decisions in flowcharts?**
 - a) Diamond
 - b) Rectangle
 - c) Circle
 - d) Triangle

- 3. What is the purpose of decision making in algorithms?**
 - a) Add randomness
 - b) Make longer
 - c) Control flow with conditions
 - d) Add complexity

- 4. Which shape shows input/output in a flowchart?**
 - a) Rectangle
 - b) Circle
 - c) Diamond
 - d) Parallelogram

- 5. What term describes a choice in an algorithm?**
 - a) Procedure
 - b) Action
 - c) Operation
 - d) Condition

6. What is the function of the ‘IF’ statement in algorithms?

- a) Conditional actions
- b) Create loops
- c) Add complexity
- d) Add randomness

7. Which shape shows process/action in flowcharts?

- a) Diamond
- b) Rectangle
- c) Diamond Circle
- d) Triangle

8. What are algorithms in computers?

- a) Step-by-step recipes
- b) Games
- c) Online shops
- d) Playlists

9. What is the purpose of an algorithm?

- a) Solve systematically
- b) Random steps
- c) Just equations
- d) Hardware

10. What is the typical result of evaluating a condition in an algorithm?

- a) Symbol
- b) Random number
- c) Process
- d) True/False or Yes/No

Answer:

1b	2a	3c	4d	5d	6a	7b	8a	9a	10d
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Lesson 15: Repeating in Algorithms

1. What is iteration in algorithms?

- a) Decision
- b) Selection
- c) Repeated instructions
- d) Equation

2. What is the purpose of loops?

- a) Add complexity
- b) Automate repeated tasks
- c) Make longer
- d) Remove decisions

3. What does a loop keep doing?

- a) Execute actions until condition met
- b) Add complexity
- c) Do random ops
- d) Stop algorithm

4. What is the main advantage of loops?

- a) Allow parallel tasks
- b) Add complexity
- c) Remove conditions
- d) Save time & automate

5. When does a loop stop?

- a) When condition is met
- b) After one repeat
- c) When computer off
- d) End of day

6. How can flow of a loop be controlled?

- a) Use equations
- b) Remove conditions
- c) Set start, condition, step
- d) Add more actions

7. What do loops mainly solve?

- a) Problems w/o condition
- b) Long problems
- c) Complex problems
- d) Repetition problems

8. What is the outcome of loop condition?

- a) True/False
- b) Random number
- c) Equation
- d) Action

9. What is the purpose of loops in general?

- a) Make long
- b) Automate repetition
- c) Remove decision
- d) Add randomness

10. Which term allows repeating tasks?

- a) Looping
- b) Decision
- c) Selection
- d) Sorting

Answer:

1c	2b	3a	4d	5a	6c	7d	8a	9b	10a
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Lesson 16: Using Algorithms to Sort Data

1. What is Bubble Sort used for?

- a) Make bubbles
- b) Solve equations
- c) Sort numbers/items
- d) Draw patterns

2. What does Selection Sort do?

- a) Randomly rearranges
- b) Always picks largest
- c) Picks random items
- d) Puts smallest to front

3. How does Bubble Sort work?

- a) Rearranges randomly
- b) Pops bubbles
- c) Counts items
- d) Swaps adjacent if wrong

4. What happens if there are no swaps during a pass?

- a) Done, stop
- b) Start again
- c) Keep checking forever
- d) Switch algorithm

5. What is the default order used in Bubble Sort?

- a) Random
- b) High → Low
- c) Alphabetical
- d) Low → High

6. Advantage of sorted data?

- a) Needs Bubble Sort
- b) Uses more memory
- c) Easier and faster searching
- d) Only works with numbers

7. When are two items swapped in Bubble Sort?

- a) Both even
- b) Left smaller
- c) Both prime
- d) Right smaller

8. Selection Sort vs Bubble Sort?

- a) Selection picks smallest
- b) Bubble selects largest
- c) Bubble moves to sorted
- d) Selection compares neighbors

9. Main purpose of sorting?

- a) Organize data
- b) Make random data
- c) Math work
- d) Write reports

10. Which is generally more efficient?

- a) Bubble
- b) Depends on data
- c) Same always
- d) Selection

Answers:

1c	2d	3d	4a	5d	6c	7d	8a	9a	10b
----	----	----	----	----	----	----	----	----	-----

Lesson 17: Understanding and Development of Algorithms

1. Purpose of a search algorithm?

- a) Count items
- b) Arrange order
- c) Find a value
- d) Organize categories

2. Which is a search algorithm?

- a) Adding
- b) Sorting
- c) Looping
- d) Linear Search

3. What is another name for a linear search?

- a) Binary Search
- b) Sorting Search
- c) Serial Search
- d) Random Search

4. In linear search, how is value checked?

- a) Only last
- b) Random
- c) Each from start to end
- d) Only first

5. What must be done before binary search?

- a) Make new list
- b) Count list
- c) Sort ascending
- d) Find middle

6. Where does binary search start?

- a) Beginning
- b) Middle
- c) End
- d) Random

7. What is the process of dividing a list into halves called?

- a) Binary Chop
- b) Binary Split
- c) Value Swap
- d) Data Shuffle

8. If target > middle in binary search?

- a) Ignore first half
- b) Ignore none
- c) Ignore second half
- d) Ignore both

9. Why binary search for large sets?

- a) Faster than linear
- b) Works unsorted
- c) Fewer comparisons
- d) Always finds value

10. Why data sorted for binary search?

- a) Correct & efficient
- b) Guarantee found
- c) Make bigger
- d) Look neat

Answers:

1c	2d	3c	4c	5c	6b	7a	8a	9c	10a
----	----	----	----	----	----	----	----	----	-----

Lesson 18: Understanding Decomposition and Abstraction

1. What is decomposition?

- a) Add complexity
- b) Break into parts
- c) Hide details
- d) Program type

2. What is abstraction?

- a) Keep essentials
- b) Add complexity
- c) Hide details
- d) Detailed view

3. Example of abstraction?

- a) High-level summary
- b) Complex code
- c) Blueprint
- d) Map with all details

4. Why is decomposition useful?

- a) Hide details
- b) Add confusion
- c) Manageable parts
- d) Make complex

5. How often is decomposition used?

- a) Rarely
- b) Sometimes
- c) Only hard cases
- d) Often

6. Main goal of abstraction?

- a) Hide unneeded
- b) Focus essentials
- c) Hide relevant
- d) Complicate

7. Which can be decomposed?

- a) Sleeping
- b) Making sandwich
- c) Breathing
- d) Dreaming

8. Which is NOT abstraction?

- a) Recipe
- b) Traffic light red
- c) Car diagram
- d) Movie summary

9. What does ‘hide complexity’ mean?

- a) Simplify use
- b) Add junk
- c) Make hard
- d) Remove all details

10. Which problem needs decomposition?

- a) Simple math
- b) Writing a word
- c) Counting fingers
- d) Designing a game

Answers:

1b	2a	3a	4c	5d	6b	7b	8b	9a	10d
----	----	----	----	----	----	----	----	----	-----

Lesson 19: Problem Solving Using Decomposition

1. Decomposition means?

- a) Break into parts
- b) Create new
- c) Make harder
- d) Combine problems

2. First step in decomposition?

- a) Ignore issue
- b) Combine parts
- c) Solve at once
- d) Split into parts

3. Good analogy of decomposition?

- a) Writing
- b) Jigsaw puzzle
- c) Baking cake
- d) Maze

4. How does decomposition help?

- a) New problems
- b) Makes harder
- c) Smaller parts
- d) Solve at once

5. Key benefit of using operators?

- a) Simplifies code and logic
- b) Adds unnecessary complexity
- c) Makes programs harder to read
- d) Combines unrelated elements

6. What does solving a small part mean?

- a) Final solution
- b) Combine
- c) Solve that part
- d) Ignore

7. Final step in decomposition?

- a) Break parts
- b) Combine solutions
- c) Solve smallest
- d) Ignore

8. NOT decomposition?

- a) Merge into one big problem
- b) Solve parts separately
- c) One part at a time
- d) Break problem

9. Why do we break problems into smaller parts?

- a) Make it more complicated
- b) Combine unrelated things
- c) Simplify the problem
- d) Ignore parts

10. Decomposition allows a task to be completed as:

- a) Fastest
- b) One whole
- c) Team
- d) Small parts

Answers:

1a	2d	3b	4c	5a	6c	7b	8a	9c	10d
----	----	----	----	----	----	----	----	----	-----

Lesson 20: Problem Solving Using Decomposition (Vending Machine)

1. What is the first step in solving a vending machine problem?

- a) Remove coins
- b) Check buttons
- c) Identify function
- d) Clean slot

2. What does speed mean when evaluating an algorithm?

- a) Troubleshoot
- b) Cleaning
- c) Decomposition
- d) Coin removal

3. Which is abstraction in vending machine?

- a) Coin check
- b) Maintenance
- c) Dispense part
- d) User interface

4. After solving a part, next?

- a) Retest
- b) Call help
- c) Clean slot
- d) Find another

5. Why flowchart in decomposition?

- a) Skip steps
- b) Avoid work
- c) Show process
- d) Complicate

6. What does decomposition mean in the context of a vending machine?

- a) Split functions
- b) Ignore
- c) Merge problems
- d) Break machine

7. Benefit of decomposition?

- a) Harder
- b) Simplify
- c) Merge unrelated
- d) Add complex

8. What is the main goal of comparing algorithms?

- a) To clean the slot again
- b) To call for technical help
- c) To identify which algorithm is most efficient or effective
- d) To ignore the problem and move on

9. Final step in solving?

- a) Fix all at once
- b) Clean slot
- c) Ignore
- d) Combine fixes

10. Why is decomposition useful?

- a) Complicate
- b) No planning
- c) Hide details
- d) Focus parts

Answers:

1c	2a	3d	4a	5c	6a	7b	8c	9d	10d
----	----	----	----	----	----	----	----	----	-----

Lesson 21: Comparing and Evaluating Algorithms

1. Important factors in evaluation?

- a) Shape, texture
- b) Speed, simplicity
- c) Color, size
- d) Sound, taste

2. What does speed mean when evaluating an algorithm?

- a) Fast finish
- b) Complex
- c) Steps
- d) Colorful

3. Simplicity means?

- a) Steps
- b) Easy to understand
- c) Colorful
- d) Fast

4. Best algorithm is?

- a) Fast + simple
- b) Runs slow
- c) Colorful
- d) Most steps

5. What is an algorithm?

- a) Computer type
- b) Complex equation
- c) Flowchart type
- d) Step-by-step method

6. Why compare solutions?

- a) Avoid work
- b) Make complex
- c) Find best
- d) Pick worst

7. How does evaluation help?

- a) Add complexity
- b) Compare methods
- c) Focus speed only
- d) Ignore others

8. What is the main goal of comparing algorithms?

- a) Most efficient
- b) Colorful
- c) Most steps
- d) Slowest

9. NOT a factor in evaluation?

- a) Speed
- b) Effectiveness
- c) Color
- d) Simplicity

10. Goal when comparing?

- a) Slowest
- b) Best for task
- c) Involves the most steps
- d) Most complex

Answers:

1b	2a	3b	4a	5d	6c	7b	8a	9c	10b
----	----	----	----	----	----	----	----	----	-----

Lesson 22: Problem Solving Using Algorithms

1. What should you do before choosing an algorithm?

- a) Random pick
- b) Ask teacher
- c) Start coding
- d) Understand problem

2. Time efficiency means?

- a) Runs fast
- b) Colorful
- c) Steps
- d) Complex

3. Why accuracy important?

- a) Add detail
- b) Solve right
- c) More steps
- d) Look good

4. Scalability means?

- a) Step count
- b) Works small + large
- c) Only small
- d) Colorful

5. Goal in choosing algorithm?

- a) Solve well + fast
- b) Visuals
- c) Longest
- d) Complicate

6. First step in selection?

- a) Test data
- b) Code once
- c) Understand
- d) Think twice

7. Why break into parts?

- a) Complicate
- b) Confuse
- c) Avoid
- d) Understand better

8. Factor for large tasks?

- a) Speed
- b) Popular
- c) Complexity
- d) Color

9. Efficiency means?

- a) Runs fast
- b) Complexity
- c) Color
- d) Resources

10. NOT a factor in evaluating?

- a) Resources
- b) Time
- c) Accuracy
- d) Color

Answers:

1d	2a	3b	4b	5a	6c	7d	8a	9a	10d
----	----	----	----	----	----	----	----	----	-----

Lesson 23: Storing and Executing Instruction on a Computer

1. What is the CPU commonly called?

- a) Microprocessor
- b) Main chip
- c) Processor
- d) Hard drive

2. What is the main role of the CPU?

- a) Printing docs
- b) Storing data
- c) Processing + executing
- d) Displaying graphics

3. What defines a program?

- a) Physical parts
- b) Instructions for a task
- c) Virus type
- d) Power source

4. Before CPU use, where are instructions loaded?

- a) Hard drive
- b) Keyboard
- c) Monitor
- d) Main memory

5. Role of the Control Unit (CU)?

- a) Decodes instructions
- b) Stores memory
- c) Runs arithmetic
- d) Controls graphics

6. Which part does arithmetic/logic?

- a) Control Unit
- b) Main memory
- c) Hard drive
- d) ALU

7. Another name for registers?

- a) Hard drive
- b) High-speed CPU storage
- c) Main memory
- d) Control Unit

8. Why does CPU use registers?

- a) Large storage
- b) Arithmetic
- c) Memory mgmt
- d) Fast small storage

9. Fetch-execute cycle is?

- a) CPU gets instructions + executes
- b) DB fetch + query
- c) Internet fetch + run
- d) executes Web fetch + run

10. What carries addresses between CPU and memory?

- a) Control Bus
- b) Address Bus
- c) Data Bus
- d) Decode Bus

Answer Strip:

1c	2c	3b	4d	5a	6d	7b	8d	9a	10b
----	----	----	----	----	----	----	----	----	-----

Lesson 24: Representing Data in Binary Format

1. ASCII stands for?

- a) American Standard Code for Info Interchange
- b) Alphabetical Symbol Code
- c) Advanced System Character ID
- d) All System Characters Input Code

2. How many bits are used to represent an ASCII character?

- a) 10
- b) 6
- c) 7
- d) 8

3. Why ASCII?

- a) Letters/numbers in binary
- b) Decimal data
- c) Hex data
- d) Octal data

4. Bits in a Byte?

- a) 10
- b) 8
- c) 6
- d) 4

5. Which is NOT ASCII?

- a) Chinese characters
- b) Upper-case
- c) Lower-case
- d) Numbers

6. One hex digit can show?

- a) 32
- b) 16
- c) 8
- d) 4

7. Unicode means?

- a) Uniform Code for Internet Docs
- b) Universal Character Encoding Standard
- c) Unique Code for Data
- d) Universal Info Display

8. Basic Unicode bits per char?

- a) 32
- b) 8
- c) 16
- d) 4

9. Hex system base?

- a) 16
- b) 2
- c) 10
- d) 8

10. Binary 1011 to decimal?

- a) 10
- b) 6
- c) 8
- d) 11

Answer Strip:

1a	2c	3a	4b	5a	6b	7b	8c	9a	10d
----	----	----	----	----	----	----	----	----	-----

Lesson 25: Representing and Manipulating Different Data Types

1. Purpose of different image formats?

- a) Reduce quality
- b) Improve look
- c) Store differently
- d) Confuse users

2. Which image format compresses efficiently with minimal quality loss?

- a) PNG
- b) GIF
- c) BMP
- d) TIFF

3. File extension for JPEG?

- a) jpg
- b) bmp
- c) png
- d) gif

4. Colors in 16-color bitmap?

- a) 32
- b) 128
- c) 256
- d) 16

5. Format for line drawings?

- a) TIFF
- b) PNG
- c) BMP
- d) GIF

6. GIF stands for?

- a) Graphic Image Format
- b) General Image File
- c) Graphics Interchange Format
- d) Graphics Internal Form

7. Which factor primarily affects file size when converting JPEG to BMP?

- a) Image resolution
- b) File name length
- c) Number of folders
- d) Color of background

8. Bytes per pixel in B/W image?

- a) 2.5
- b) 1
- c) 0.125
- d) 3

9. Bytes per pixel in RGB?

- a) 4
- b) 2
- c) 3
- d) 1

10. Which is NOT a primary screen color?

- a) Blue
- b) Red
- c) Yellow
- d) Green

Answer:

1c	2a	3a	4d	5b	6c	7a	8c	9c	10c
----	----	----	----	----	----	----	----	----	-----

Lesson 26: Designing Computer Programs

1. What is the goal of designing a computer program?

- a) Plan steps for computer
- b) Make OS
- c) Organize files
- d) Make PC look nice

2. Which is NOT a tool for design?

- a) Creativity
- b) Problem solving
- c) Keyboard shortcuts
- d) Logical thinking

3. Why logic needed?

- a) Faster program
- b) Break problems
- c) Add color
- d) Add creativity

4. In Scratch, green flag does?

- a) Save
- b) Delete
- c) Start program
- d) Change sprite color

5. Role of creativity?

- a) Make longer code
- b) Write foreign code
- c) Strict rules
- d) Explore approaches

6. NOT in design step?

- a) Memorize code
- b) Testing
- c) Creativity
- d) Logic

7. Flowchart used to?

- a) Write code
- b) Compile code
- c) Show logic visually
- d) Execute program

8. “Execute” means?

- a) Compile
- b) Design
- c) Write
- d) Run code

9. Why pseudocode?

- a) Plain English code
- b) Faster run
- c) Animation
- d) Foreign code

10. Main goal of coding?

- a) Pretty PC
 - b) Step-by-step instructions
 - c) Animations
 - d) Organize files
- Answer Strip:
- i) Answer:

1a	2c	3b	4c	5d	6a	7c	8d	9a	10b
----	----	----	----	----	----	----	----	----	-----

Lesson 27: Designing Computer Programs

1. Which of the following are the three main elements of an algorithm?

- a) Sequencing, iteration, selection
- b) Loops, variables, functions
- c) Colors, shapes, sizes
- d) Sequencing, iteration, multiplication

2. Selection in programs is?

- a) Condition-based choice
- b) Stop program
- c) Text color
- d) Add comments

3. Pseudocode is?

- a) Foreign language code
- b) Works only old PCs
- c) Slows PC
- d) Plain English code

4. What is a loop?

- a) Programming language
- b) Variable type
- c) Repeat until condition
- d) Cable knot

5. Sequencing means?

- a) Repeating
- b) Decisions in code
- c) Steps in order
- d) Colorful output

6. Another word for selection?

- a) Iteration
- b) Conditional
- c) Abstraction
- d) Multiplication

7. Iteration vs sequencing?

- a) Unrelated
- b) Sequencing repeats, iteration orders
- c) Same
- d) Iteration repeats, sequencing order

8. Why use nesting + random?

- a) Store in DB
- b) Ask same Q with different numbers
- c) Display order
- d) Lengthen code

9. Purpose of pseudocode?

- a) Faster code
- b) Plan in plain English
- c) Animation
- d) Confuse coders

10. Repetition of instructions is?

- a) Abstraction
- b) Iteration
- c) Sequencing
- d) Selection

Answer Strip:

1a	2a	3d	4c	5c	6b	7d	8b	9b	10b
----	----	----	----	----	----	----	----	----	-----

Lesson 28: Designing Computer Programs

1. Boolean represents?

- a) String
- b) True/False
- c) List
- d) Decimal

2. Which is NOT a type?

- a) List
- b) Char
- c) Currency
- d) Keyword

3. What is the role of variable name in programming?

- a) Change constants
- b) Shorten code
- c) Identify stored data
- d) Make graphics

4. Single character type?

- a) Int
- b) String
- c) List
- d) Char

5. What does 'dynamic' mean in programming?

- a) Fixed
- b) Changeable
- c) Inefficient
- d) Complex

6. Constant is?

- a) Keyword
- b) Value changes often
- c) Named variable
- d) Fixed value

7. Why constants > hard coding?

- a) Easier to update
- b) Runtime update
- c) Makes shorter
- d) Easy to recall

8. Variable is?

- a) Constant
- b) Temp data store
- c) Loop keyword
- d) Fixed value

9. Whole numbers in Python?

- a) Float
- b) Bool
- c) Int
- d) Str

10. Function of variables?

- a) Create graphics
- b) Store data
- c) Replace constants
- d) Control loops

Answer Strip:

1b	2c	3c	4d	5b	6d	7a	8b	9c	10b
----	----	----	----	----	----	----	----	----	-----

Lesson 29: Designing Computer Programs

1. Operator for equal?

- a) !=
- b) ==
- c) >=
- d) >

2. OR operator?

- a) &&
- b) !
- c) ||
- d) ==

3. What is the result of $(12 / 3) + 2$?

- a) 4
- b) 6
- c) 8
- d) 10

4. Operator for less than?

- a) <
- b) >
- c) <=
- d) >=

5. NOT operator?

- a) &&
- b) ||
- c) ==
- d) !

6. $(8 - 3) * 4 + 2 = ?$

- a) 20
- b) 22
- c) 16
- d) 24

7. $\geq$ operator?

- a) $>$
- b) $<=$
- c) $>=$
- d) $<$

8. Not equal operator?

- a) $==$
- b) $>$
- c) $<$
- d) $!=$

9. $(7 + 3) / (2 - 1) = ?$

- a) 5
- b) 10
- c) 7
- d) 4

10. AND operator?

- a) $\&\&$
- b) $\parallel$
- c) $==$
- d) $!$

Answer Strip:

1b	2c	3b	4a	5d	6b	7c	8d	9b	10a
----	----	----	----	----	----	----	----	----	-----

Lesson 30: Designing Computer Programs

1. What is the purpose of the return keyword in the programming?

- a) Send value back
- b) Import modules
- c) Define variable
- d) Declare function

2. Which NOT modular design?

- a) Encapsulation
- b) Easy bug fixing
- c) Divide into modules
- d) One huge module

3. `print("Hello, World!")` does?

- a) Asks input
- b) Displays text
- c) Defines variable
- d) Calculates sum

4. Modular design means?

- a) No functions
- b) Random many modules
- c) Divide into smaller modules
- d) Furniture design

5. Result of `"5" + "3"`?

- a) 53
- b) 15
- c) 35
- d) 8

6. Parameter passing is?

- a) Sectioning code
- b) Giving data to function
- c) Color change
- d) Speed up code

7. Python is?

- a) Snake
- b) Equation
- c) Software
- d) Programming language

8. def keyword does?

- a) Declare function
- b) Define variable
- c) Import
- d) Print

9. Advantage of modular design?

- a) Harder to reuse
- b) Easier maintenance
- c) Longer code
- d) Less reusability

10. import in Python does?

- a) Include modules
- b) Print output
- c) Make variable
- d) Define function

Answer:

1a	2d	3b	4c	5a	6b	7d	8a	9b	10a
----	----	----	----	----	----	----	----	----	-----

j)